MHF4U Advanced Functions - Unit 1 Practice Test, Version 3

Worked key and thinking guide

50 marks. The category labels on the student paper identify what each part assesses. Award equivalent correct reasoning. Graphs require labelled key points and correct open or closed endpoints.

1. Piecewise functions [9]

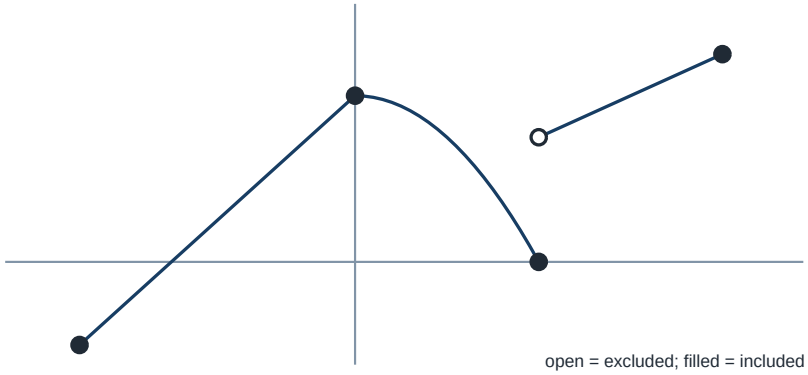
a. Draw y=2x+4 from filled (-3,-2) toward open (0,4); draw y=4-x^2 from filled (0,4) through (1,3) to filled (2,0); draw y=x+1 for 2<x<=4, with open (2,3) and filled (4,5). [4]

b. Domain [-3,4]; range [-2,5]. [2]

c. The first branch increases from x=-3 to x=0, and the third branch increases for 2<x<=4. Accept [-3,0] and (2,4] as branch intervals. [1]

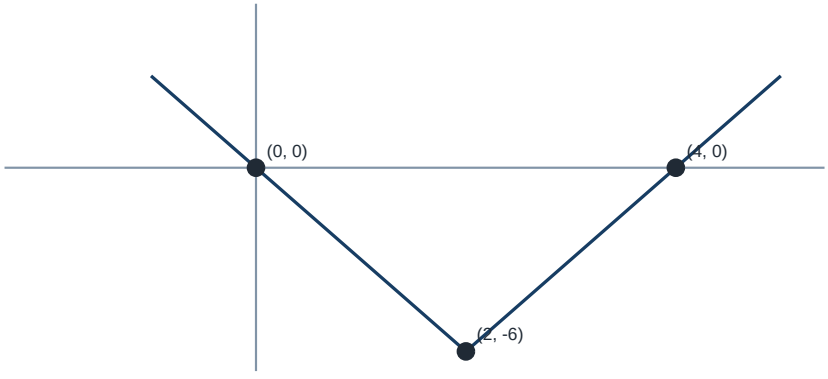
d. No. A horizontal line such as y=3 meets the graph more than once: S(-0.5)=3 and S(1)=3. The inverse would give input 3 two outputs. [2]

Thinking move: Test horizontal lines on the original graph. Repeated outputs become repeated inputs in the inverse.



2. Absolute value graph [6]

The vertex is (2,-6) and the V opens up. Solve 3|x-2|-6=0, so |x-2|=2 and x-intercepts are (0,0) and (4,0). Label these on the sketch. [3] Domain (-infinity,infinity); range [-6,infinity). [2] The coefficient 3 stretches the parent V vertically by factor 3; there is no reflection. [1]



3. Transforming a function [5]

a. Match the original input 9: 4(x+2)=9, hence x=1/4. The new output is -2(3)+5=-1, so the point is (1/4,-1). [2]

b. The square-root input must satisfy 4(x+2)>=0, so domain [-2,infinity). Since f is nonnegative, h<=5; range (-infinity,5]. [2]

c. Horizontally compress the graph by factor 1/4, then shift it left 2. [1]

Thinking move: Set the whole inside expression equal to the old x-coordinate before finding the new point.

4. Inverse relations [5]

a. Original q has domain all reals and range `[2,infinity)`. The inverse relation therefore has domain `[2,infinity)` and range `(-infinity,infinity)`. [2]

b. The parabola fails the horizontal-line test: for example `q(2)=3=q(4)`, so the inverse would send input 3 to two outputs. [1]

c. One valid restriction is `x>=3`; swap x and y in `y=(x-3)^2+2`, then take the positive root to obtain `q^(-1)(x)=3+sqrt(x-2)`, domain `x>=2`. Accept the alternative restriction `x<=3` and inverse `3-sqrt(x-2)`. [2]

Thinking move: Restrict to one side of the vertex. That choice determines which square-root branch is valid.

5. Observation wheel diagram [7]

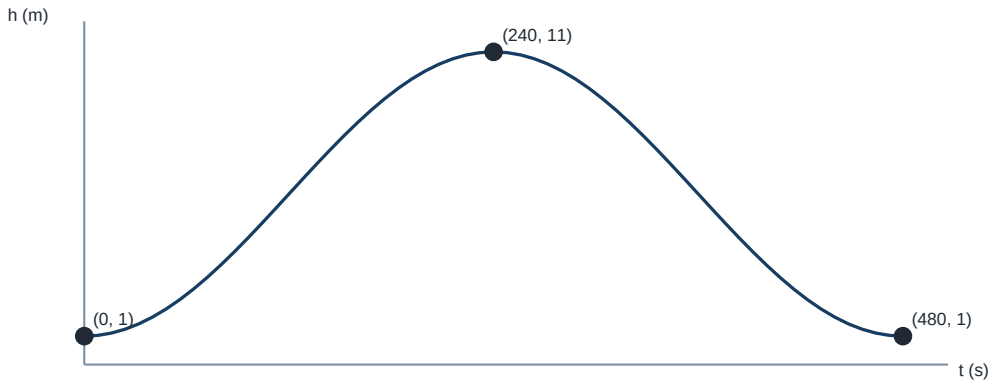
a. Centre height `1+5=6 m`; greatest seat height `6+5=11 m`. [2]

b. Domain `[0,480]` seconds; range `[1,11]` metres. [2]

c. Six metres is the height of the centre, not the top. Add the radius once more: `6+5=11 m`. [1]

d. Draw one smooth cycle starting at `(0,1)`, rising to `(240,11)`, and falling to `(480,1)`. Label the axes time (s) and height (m). [2]

Thinking move: Minimum = ground clearance; centre = minimum + radius; maximum = centre + radius. Half a rotation takes half the time.



6. Average and instantaneous rates [8]

a. `H(0)=2` and `H(2)=26`; average rate `(26-2)/(2-0)=12 m/s`. [2]

b. `H(1.9)=25.37` and `H(2.1)=26.57`; centred estimate `(26.57-25.37)/(2.1-1.9)=6 m/s`. [3]

c. Complete the square: `H(t)=-3(t-3)^2+29`. The maximum is `29 m` at `t=3 s`. [2]

d. The first rate averages motion over two seconds. The second estimates the rate only near `t=2`; the ball's upward speed changes during the interval. [1]

Thinking move: A secant slope describes an interval; a tangent estimate describes one instant. Include height-per-time units.

7. Graph properties and interval notation [5]

a. `r(-x)=(-x)^4-3(-x)^2=x^4-3x^2=r(x)`, so `r` is even. [2]

b. Y-axis symmetry gives `(-2,4)`. [1]

c. `|x+2|<4` means `-4<x+2<4`, so `-6<x<2`, or `(-6,2)`. [2]

8. Secant and tangent diagram [5]

a. Line AB is the secant and represents the average rate: `(6-2)/(4-2)=2` units of y per unit of x. [2]

b. The horizontal line through A is tangent at A and represents the instantaneous rate there; slope `0`. [2]

c. The secant measures the whole interval from 2 to 4, where the curve rises. The tangent describes the curve's direction only at the minimum A, where it is flat. [1]

Thinking move: Average rate uses a secant through two x-values. Instantaneous rate uses a tangent at one x-value.